

host_agent

All mobile homes and caravan homes in Exeter have been mapped for you.

- 212 mobile home buildings and 212 building parts were found and plotted.
- The map includes:
- Green: Mobile home buildings
- Orange: Mobile home building parts
- Blue: Exeter city boundary

You can view all the locations and details on the interactive map: [mobile_homes_exeter_map.html](#)

If you need further filtering or want to see details for specific areas or types, let me know!

human

Show me all the mobile homes or caravans homes in Exeter.?

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are mobile homes or caravan homes, and we need to find as many entries as possible for these entities in Exeter.

output steps: ["Find Exeter 1 area", "Search for mobile homes or caravan homes in Exeter as many search results"]

named_area_agent

The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This GeoDataFrame contains only the polygon(s) representing Exeter, filtered by name and relevant settlement descriptions.

data_analysis_agent

Filtered to 1 polygon for Exeter using the name and description columns. The result is saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) representing Exeter, filtered by name and relevant settlement descriptions. If you need to view or analyze this polygon further, let me know!

land_use_features_agent

No caravan parks or mobile home parks were found within the 'filtered_exeter_area_polygon'. If you have any other queries or need further assistance, feel free to ask!

buildings_agent

Search Results: - 69,169 mobile homes or caravan homes found in the 'filtered_exeter_area_polygon' (artifact: buildings_mobile_homes_exeter_area). - 85,480 building parts related to mobile homes or caravan homes found (artifact: buildingpart_mobile_homes_exeter_area).

Artifacts generated: - buildings_mobile_homes_exeter_area: Contains building data for mobile homes/caravan homes within the specified area. - buildingpart_mobile_homes_exeter_area: Contains building part data for mobile homes/caravan homes within the specified area.

If you need further filtering or analysis, please specify your requirements!

plotting_agent

Plotted 212 mobile home buildings and 212 building parts in Exeter.

First 5 buildings: - {'osid': '4517d436-2e23-4168-9f0a-113446a29941', 'description': 'Static Caravan Or Mobile Home', 'geometry_area_m2': 91.521, ...} - {'osid': 'faacd2d4-a9eb-49c2-88b2-178a0eabe119', 'description': 'Static Caravan Or Mobile Home', 'geometry_area_m2': 58.02, ...} - {'osid': '7bf4d653-70bf-4ade-b56e-41be6f4216b9', 'description': 'Static Caravan Or Mobile Home', 'geometry_area_m2': 60.344, ...} - {'osid': '8e0ac9e0-0d52-4754-982a-f6b8eea5d08d', 'description': 'Static Caravan Or Mobile Home', 'geometry_area_m2': 78.247, ...} - {'osid': '9d0d2051-5ba5-408d-9d38-9cc7fd4264dc', 'description': 'Static Caravan Or Mobile Home', 'geometry_area_m2': 94.848, ...}

First 5 building parts: - {'osid': '1a312315-e99e-44a0-bc4d-1f64f12d68a9', 'description': 'Static Caravan Or Mobile Home', 'geometry_area_m2': 78.247, ...} - {'osid': 'ec0eaf75-be59-4cc3-8146-6cdccb7772c1', 'description': 'Static Caravan Or Mobile Home', 'geometry_area_m2': 94.848, ...} - {'osid': 'c531be46-49d8-4010-a1f1-bb3d355eb325', 'description': 'Static Caravan Or Mobile Home', 'geometry_area_m2': 81.605, ...} - {'osid': 'd0a22ace-fdf7-476b-98a8-4d06e1e03d3f', 'description': 'Static Caravan Or Mobile Home', 'geometry_area_m2': 84.073, ...} - {'osid': '8db4a9ff-1565-4e13-a5f5-56de39ce545a', 'description': 'Static Caravan Or Mobile Home', 'geometry_area_m2': 78.322, ...}

An interactive map has been generated showing all mobile and caravan homes in Exeter: - Green: mobile home buildings - Orange: mobile home building parts - Blue: Exeter city boundary

Artifact name: mobile_homes_exeter_map.html

Description: Folium map showing all mobile and caravan homes in Exeter (green: buildings, orange: building parts, blue: Exeter boundary)

File: mobile_homes_exeter_map.html

Tool Calls and Code

Tool: send_message

- Args: {'target': 'planning_agent', 'task_description': 'User wants to see all mobile homes or caravan homes in Exeter. Provide the general steps to solve this geospatial query.'}
- Response: The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are mobile homes or caravan homes, and we need to find as many entries as possible for these entities in Exeter.

output steps: ["Find Exeter 1 area", "Search for mobile homes or caravan homes in Exeter as many search results"]

Tool: send_message

- Args: {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}
- Response: The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This GeoDataFrame contains only the polygon(s) representing Exeter, filtered by name and relevant settlement descriptions.

Tool: call_os_ngd

- Args: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}

- Response: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..., count of records fetched : 98.

Tool: send_message

- Args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}
- Response: Filtered to 1 polygon for Exeter using the name and description columns. The result is saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) representing Exeter, filtered by name and relevant settlement descriptions. If you need to view or analyze this polygon further, let me know!

Tool: generate_metadata_for_artifacts

- Args: {'artifact_names': ['exeter_area_polygon']}

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter')) &
                    (gdf['description'].str.lower().isin(['city', 'town', 'suburban area', 'part of settlement', 'village', 'settlement'])))]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and description columns."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and relevant settlement descriptions.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool: send_message

- Args: {'target': 'land_use_features', 'task_description': "Search for caravan parks or mobile home parks within 'filtered_exeter_area_polygon'."}
- Response: No caravan parks or mobile home parks were found within the 'filtered_exeter_area_polygon'. If you have any other queries or need further assistance, feel free to ask!

Tool: send_message

- Args: {'target': 'buildings', 'task_description': "Search for mobile homes or caravan homes within 'filtered_exeter_area_polygon'."}
- Response: Search Results:
- 69,169 mobile homes or caravan homes found in the 'filtered_exeter_area_polygon' (artifact: buildings_mobile_homes_exeter_area).
- 85,480 building parts related to mobile homes or caravan homes found (artifact: buildingpart_mobile_homes_exeter_area).

Artifacts generated: - buildings_mobile_homes_exeter_area: Contains building data for mobile homes/caravan homes within the specified area. - buildingpart_mobile_homes_exeter_area: Contains building part data for mobile homes/caravan homes within the specified area.

If you need further filtering or analysis, please specify your requirements!

Tool: generate_metadata_for_all_artifacts

- Args: {}

Tool: send_message

- Args: {'target': 'plotting_agent', 'task_description': "Plot all mobile homes or caravan homes in Exeter using the artifacts 'buildings_mobile_homes_exeter_area' and 'buildingpart_mobile_homes_exeter_area' within the area polygon 'filtered_exeter_area_polygon'."}

Tool: generate_metadata_for_artifacts

- Args: {'artifact_names': ['buildings_mobile_homes_exeter_area', 'buildingpart_mobile_homes_exeter_area', 'filtered_exeter_area_polygon']}

Tool Code Executor

```
def plot_mobile_homes_exeter(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Unpack data
    gdf_buildings = data[0]
    gdf_buildingparts = data[1]
    gdf_exeter = data[2]

    # Ensure all CRS are EPSG:4326 for folium
    gdf_buildings = gdf_buildings.to_crs(epsg=4326)
    gdf_buildingparts = gdf_buildingparts.to_crs(epsg=4326)
    gdf_exeter = gdf_exeter.to_crs(epsg=4326)
```

```
# Get the Exeter polygon
```

```
exeter_poly = gdf_exeter.iloc[0].geometry
```

```
# Filter buildings and buildingparts that are within Exeter
```

```
gdf_buildings_in_exeter = gdf_buildings[gdf_buildings.geometry.within(exeter_poly)]
```

```
gdf_buildingparts_in_exeter = gdf_buildingparts[gdf_buildingparts.geometry.within(exeter_poly)]
```

```
# Create folium map centered on Exeter
```

```
centroid = exeter_poly.centroid
```

```
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)
```

```
# Add Exeter boundary
```

```
folium.GeoJson(mapping(exeter_poly), name='Exeter Boundary', style_function=lambda x: {'color': 'blue', 'weight': 2, 'fillOpacity': 0.05}).add_to(m)
```

```
# Plot mobile homes from buildings
```

```
for _, row in gdf_buildings_in_exeter.iterrows():
```

```
    folium.GeoJson(row.geometry,
```

```
                    style_function=lambda x: {'color': 'green', 'weight': 2, 'fillOpacity': 0.3},
```

```
                    tooltip=f"Building: {row['description']} | Area: {row['geometry_area_m2']} m2").add_to(m)
```

```
# Plot mobile homes from buildingparts
```

```
for _, row in gdf_buildingparts_in_exeter.iterrows():
```

```
    folium.GeoJson(row.geometry,
```

```
                    style_function=lambda x: {'color': 'orange', 'weight': 2, 'fillOpacity': 0.3},
```

```
                    tooltip=f"Building Part: {row['description']} | Area: {row['geometry_area_m2']} m2").add_to(m)
```

```
# Save map
```

```
map_filename = 'mobile_homes_exeter_map.html'
```

```
m.save(map_filename)
```

```
# Prepare summary
```

```
summary_rows = gdf_buildings_in_exeter.head(5).to_dict('records')
```

```
summary_rows_parts = gdf_buildingparts_in_exeter.head(5).to_dict('records')
```

```
summary = f"Plotted {len(gdf_buildings_in_exeter)} mobile home buildings and {len(gdf_buildingparts_in_exeter)} building parts in Exeter. First 5 buildings
```

```
return [summary, map_filename, "Folium map showing all mobile and caravan homes in Exeter (green: buildings, orange: building parts, blue: Exeter boundary)"]
```